

Modeling Energy Consumption of Small Drones for Swarm Missions

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Abstract—Drones, particularly when deployed in swarms, hold immense potential for various applications, such as aerial imaging, delivery services, disaster response, and advanced surveillance. Their effective and efficient use, however, hinges on the accurate estimation of energy consumption. This study focuses on determining energy consumption patterns for small drones weighing less than 2 kg and with diameters under 1 m. Through an extensive series of test flights, empirical energy consumption data was collected and used to develop four distinct energy models: a theoretical model based on fundamental flight forces, a simple linear regression model, a cubic polynomial regression model, and a machine learning-based model using the XGBoost Regressor algorithm. A cost function for swarm topology control was also derived for assessing energy consumption during various activities, like connectivity restoration and formation change, facilitating more energy-efficient decision-making in swarm operations. Our findings highlighted an energy-efficient “valley” within the airspeed range, indicating that flying at speeds outside this range results in higher energy consumption. The machine learning model demonstrated superior accuracy within its training range, achieving an R^2 of 0.9999, whereas the polynomial regression model was deemed best for extrapolation purposes, delivering an R^2 of 0.966. Simple linear regression and theoretical models, although less accurate, can offer quick energy demand calculations and insights into the effects of hardware modifications, respectively.

Index Terms—Drone, energy model, machine learning, regression, swarm, UAV, XGBoost.

I. INTRODUCTION

DRONES, as a subclass of unmanned aerial vehicles (UAV), is among several technologies that are shaping the 21st century. They are anticipated to be very useful in wildlife observation, border surveillance, wireless network provision, search and rescue missions, delivery tasks, military operations, and many other scenarios [1], [2], [3], [4], [5].

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Yet what elevates the drones’ efficiency in such missions to a different level is the development of autonomous fleets of networked drones, also called swarms.

Drones are mostly susceptible to critical failures, battery outages, physical attacks [6], and accidental impacts. In such cases, the complete loss of a drone is common, and this may potentially cause major disconnections in the swarm’s wireless network. Thus, a current research and development challenge is to enable swarms with the ability to restore their (k -) connectivity by relocating one or more drones in case of the aforementioned incidents [7]. However, without thorough consideration, such topology maintenance activities can be a burden to the swarm in terms of energy consumption. In order to determine the optimum selection of the topology to be formed, drone(s) to be relocated, and routes to be taken, the energy consumption characteristics of drones must be modeled as a cost function. This cost function will enable the evaluation of alternative topology and formation control methods and energy-optimal decision-making.

Drones are designed and produced in varying sizes, weights, and body types, as they can be useful in different missions. Regardless of their size and shape, they benefit from and are affected by similar physical phenomena. While a larger drone can carry bigger payloads and specialized equipment, a smaller drone can be used more comfortably indoors, make sharp maneuvers easier, and be used by a broader user base. Small drones are not necessarily recreational or hobbyist devices but are also developed as commercial, industrial, and military tools. Yet, there is no universal standard defining the size of drones, as regulations set different definitions in various parts of the world. A drone is considered small if it is lighter than 2 kg in Canada, 25 kg in the United States and the European Union, and 20 kg in the United Kingdom. However, our definition of a small drone extends from 500 g to 1.5 kg without payload and to 2 kg gross (i.e., with all payload included). From our observations, we anticipate significant differences in energy consumption patterns out of this range due to potential component differences, including motors, propellers, etc.

In this paper, we provide four energy consumption models for small drones and experimental validation of the energy consumption models. The major contributions of the paper to the field can be summarized as follows:

- We made a comprehensive analysis of the literature and provided a simplified theoretical model of energy consumption characteristics of small rotary-wing drones, providing a baseline.

- We built a simple linear regression-based energy consumption model for quick estimations.
- We built a high-accuracy multivariate polynomial linear regression model to make accurate energy demand projections, especially out of data ranges.
- We designed a very-high-accuracy multivariate machine learning (ML) based model using XGBoost and data preprocessing methods for estimating energy consumption. To the best of our knowledge, the design of a supervised ML approach to accurately model the energy consumption of drones using experimental real-world data is a novel attempt.
- We suggested a cost function for evaluating the efficiency of swarm connectivity restoration and topology control methods based on their energy consumption. To the best of our knowledge, the derivation of a cost function tailored for use in quantifying the efficiency of topology control practices in drone swarms is unique.
- We provided an exhaustive comparative analysis concerning the estimation and projection accuracy of different modeling approaches.

The rest of the paper is organized as follows: Section II provides a deep insight into earlier attempts at modeling the energy consumption of drones as well as the topology maintenance challenges. Section III introduces some preliminary information, including factors affecting energy consumption; Section IV presents our real-world experiments; Section V includes various approaches to derive an accurate model and an inductive way of building a cost function for swarms; Section VI presents and discusses the results; And finally, Section VII concludes the study.

II. BACKGROUND

Our provision of background focuses on introducing previous efforts in estimating the energy consumption characteristics of drones during their flights and the topology control methods developed to utilize a fleet of drones together in complex missions.

A. Energy Consumption Models for Drones

The energy consumption characteristics of drones in various use cases are not yet completely uncovered. However, earlier research by Zhang et al. [8] and Muli et al. [9] provided comprehensive surveys concerning different approaches and prior art in modeling the energy consumption of drones. Zhang et al. grouped existing models and model derivation approaches under 4 classes: (i) integrated models that utilize a single dominant parameter such as lift-to-drag ratio, (ii) force component-oriented models that account for various force components, (iii) regression-based models that apply regression to fit experimental measurements, and (iv) manufacturer data-based models that calculate vendor-provided data [8]. In this study, we introduce a new approach: supervised machine learning models that are trained by experimental flight data.

The study by D'andrea [10] is one of the earliest attempts to quantify and elucidate the power requirements of a rotary-wing drone and presents an integrated model. It is based on the

generalized Breguet range equation [11] and is solely designed to describe a steady horizontal flight behavior. Therefore, it does not consider hovering and vertical (i.e., ascending and descending) movements. The model utilizes a drone's lift-to-drag ratio as a fundamental parameter, as well as mass, airspeed, and the power transfer efficiency of the battery. It has a variant that also includes the speed of headwind. Although its assumptions seem suitable for small drones (still somewhat on the heavier side), it is not parametric as many of the factors, such as the lift-to-drag ratio and cruise speed, are fixed to a predetermined value. Furthermore, finding the correct lift-to-drag ratio for their drone can be overwhelming.

Figliozzi [12] further improved the basic model proposed by D'Andrea [10] with two major contributions: a separate additive term to consider empty returns (a situation specific to delivery missions where the drone offloads its payload at halfway) and a lift-to-drag ratio that is a function of the ground speed. Most parameters are considered per a MicroDrones MD4-3000 drone and are constant (not parameterized). The presumed drone and payload weights are slightly over our small drone conceptions. The model does not consider avionics, yet it includes a margin for undocumented negative factors that may increase power consumption (e.g., headwind) but does not define individual parameters to be calculated. This model does not support hovering either.

Xu [13] also made adjustments on [10] using their empirical data. Instead of a fixed empty weight value, they considered a coarse weight calculation to reflect the impact of power requirements driven by the size and weight of the motors used. The steady flight and hover efficiencies are computed utilizing the idealized actuator disk theory and adjusted empirically to consider non-ideal rotors. For the calculations, the motor size was chosen to reflect the worst-case limited-power conditions to achieve more realistic results. However, their drone conception is certainly not a quadcopter as it has different dedicated motors for cruising and hovering; hence, the model is not accurate for what we call small drones.

The component-based models consider the essential forces a rotary-wing aircraft must create or contend with during its flight, which consist of the aircraft's weight force (attributable to gravity) and drag forces. The drag force occurs as the aircraft moves through the atmosphere and also as a result of airflow redirection to generate the lift that sustains the aircraft in flight. Stolaroff et al. [14] provide important contributions to the research concerning the energy consumption of drones and the impact of drone delivery missions on the environment. It presents a clear interpretation of the energy consumption characteristics of rotary-wing delivery drones; however, it does not focus on modeling such energy consumption and is insufficient on its own to build a model for several reasons. First, although the discussion is based on established theoretical facts and relations, it does not come up with a multi-parametric estimation model that considers all major real-world flight and environmental factors. Likewise, the assumptions on parameters like wind speed, wind direction, and power efficiency are fairly incomplete. Yet, this study still constitutes a major reference point for our research.

Another component-oriented model was proposed by Kirchstein [15], based on a previous study by Langelaan et al. [16]. It considers a predefined delivery scenario including takeoff and ascending movement with preset parameters (e.g., angle, speed, altitude, etc.) followed by a steady flight, descending, and finally landing, again with preset parameters. There is also a return route with no payload. The model reflects energy consumption caused by different force components like induced drag, parasite drag, and rotational drag against propellers, as well as additional factors like avionics, losses, and efficiency. On the downside, the model is designed for larger drones, heavier than 10 kg and capable of carrying at least 2.5 kg payloads, which have considerably different consumption rates than their smaller counterparts. Another downside is the preset delivery scenario with constant parameters used to derive the model, which limits its flexibility and adaptability.

An alternative methodology for predicting drone energy usage involves utilizing regression models that are grounded in empirical data obtained from real flights. Tseng et al. [17] present a nine-term nonlinear regression model for drone power consumption that includes horizontal and vertical speed and acceleration, as well as payload mass and wind speed. Dorling et al. [18] built their model based on the helicopter characteristics as introduced in [19]. They assumed that the energy consumption during steady horizontal flight is approximately the same as or slightly lower than that of hovering. They suggested a linear regression model to reveal the relation between energy consumption and drone weight. Their study is suitable for small drones. Nevertheless, there are significant cases where their assumption regarding the hovering energy being the same as (or higher than) steady flight does not hold, for example, flying at very high speeds or against an excess headwind, etc., as shown in some other works [8], [14] and in ours. On the other hand, Liu et al. [20] claim that ascending requires 9.8% more power than hovering, and descending takes 8.5% less power than hovering steadily.

Tseng [17] proposes a non-linear regression model composed of nine terms to estimate a drone's power requirements. It was built using a 3DR Solo drone weighing 2 kg (carrying payloads of up to 0.5 kg) and a DJI Matrice 100 drone weighing 2.8 kg. A later study [21] extends the scope with a larger drone, DJI Matrice 600. The model considers horizontal and vertical velocities, acceleration, payload mass, and wind speed, and was claimed to be demonstrating good alignment with empirical data from field trials. In the absence of wind and during steady flight, the regression model simplifies to a three-term polynomial of airspeed and mass, $P = \beta_1 \cdot v_a + \beta_7 \cdot m^2 + \beta_9$. However, our efforts to reconstruct the model unveiled that the power projections with this model do not reflect the increasing power demand observed at speeds faster than circa 12 m/s, yet there are other discrepancies with the estimated values and experimental values, pointing out a potential mismatch in the regression coefficients.

Finally, another method to calculate the energy use of drones could be by leveraging the specifications and data supplied by the manufacturer. The rates of motors, batteries, and avionics are generally published by their vendors. Yet actual

flight performance usually is not disclosed in a systematic approach. Most of the time, the manufacturer data omits environmental conditions, leaning towards an idealized scenario. Cabuk et al. [22] have previously conducted a preliminary study of a so-called holistic energy consumption model (as a function of time) for drones and tried to parameterize the individual parts' contribution to the overall energy consumption. Although relying solely on vendor data (i.e., datasheets), the study was significant in the sense of expressing how dominant the physical movement is (i.e., hovering and horizontal flying) and how exhaustive computations can be. On the downside, their work was quite superficial as it did not consider different internal and external parameters affecting energy consumption, such as the weight and speed of a drone or wind conditions.

B. Topology Control for Drone Swarms

Topology control, namely activities concerning creating, maintaining, and renewing network topologies and adjusting node placements or formations, is a crucial task for accomplishing swarm missions with drones. For instance, countering physical attacks that aim at splitting a swarm into disconnected, less capable groups by targeting its critical nodes [6] can be mitigated using such procedures. As those efforts may constitute a lifelong and continuous process, they contribute to the energy consumption of the drones in the swarm, especially due to necessary relocations.

A k -connected network remains connected even if any $k-1$ nodes malfunction. Deployment of nodes regarding k -connectivity property with maximum coverage is a reasonable approach for providing fault tolerance and application fidelity. In [23], a deployment method for achieving 2-connectivity and maximum coverage is given for ad-hoc networks. The optimality of the approach is theoretically guaranteed for various sensing and transmission ranges. Yun et al. [24] proposed another method to ensure k -connected deployment with maximum coverage for $3 \leq k \leq 6$. Geometric graph patterns can be utilized in large networks in terms of efficient deployment methods. For this purpose, MAX-Tree provides a proper tree formation with maximum coverage for flying ad-hoc networks [25]. The pattern is significantly scalable at the same time, having various properties such as fractality and symmetry. Random deployments can be advantageous for energy conservation and secure data transfer in drone swarms. A tree-based deployment scheme ensuring maximum coverage for any node count and network area size, named CoRMAC, which outputs random trees, is given in [26].

Network connectivity restoration approaches can be divided into reactive and proactive categories [27], [28]. A reactive method starts its operation when the network becomes disconnected through node and link failures. On the other hand, a proactive method aims to provide k -connectivity by executing the restoration process when the k value is decreased. Increasing radio ranges of running nodes is a fundamental technique for k -connectivity restoration as given in [29]. Obviously, the transmission power of the nodes should be adjustable to utilize this technique. Alternative to this, a backup group is maintained to replace non-operating nodes

as in Senel et al.'s approach [30]. This technique brings the extra cost of redundant nodes at the network setup phase. Another important strategy is to use the mobility capabilities of nodes to restore k -connectivity [31]. An efficient movement-based k -connectivity approach named PINC is given in [32], where the nodes are categorized as critical and non-critical. When a critical node fails to operate, the algorithm aims to move a non-critical node to the failed node at the same time conserving the minimum cost property.

III. PRELIMINARIES

A. Problem Definition

In case of device failures, battery outages, attacks with weapons, accidents, or cyber-attacks [33], a swarm of networked drones may lose one or more of its nodes permanently. In such a situation, in order to continue its mission, the swarm must collectively execute a connectivity restoration procedure, which most likely involves maintenance or regeneration of the active network topology and physical relocation of at least one of the remaining drones unless increasing the communication range of the drones solve the issue.

Restoring the connectivity of the swarm by relocating the drones may follow one of two main strategies: (i) relocating a minimal number of drones that is usually equal to the number of drones lost due to the incident, possibly by moving greater distances as drones which are safe to move (i.e., non-critical to connectivity) may typically be found at the edges of an ad-hoc topology; or (ii) relocating a group of drones, if not all, possibly by smaller distances each as this usually allow drones closer to the destination point (i.e., where the lost nodes were present) to move without vitiating the network's connectivity.

In (i), the total time required for the connectivity procedure to converge is tightly bound by the movement speed and distance to be traveled by a single (or a few) drone, resulting in a single drone undertaking the energy cost of such a movement and the rest of the swarm assuming the cost of idly waiting (i.e., hovering) for the connection to be re-established. In (ii), however, the energy cost of movement, although shall be multiplied by the number of drones moving, is going to be distributed somewhat balanced among all drones, and the cost of idle time [34] assumed by the entire swarm can be significantly reduced.

The main research question here is: "What is the best strategy in a given condition, and what are the limitations of this strategy?". Our efforts concerning measurement and modeling the energy consumption of small drones in a swarm setting will help researchers to build a proper cost function for connectivity restoration, topology control/maintenance, and formation-changing scenarios.

B. Assumptions & Definitions

In the scope of this study, we do not focus on any specific mission type, such as delivery missions, but rather aim to provide a more general point of view, unlike earlier studies [8], [15]. In addition to that, we do not use energy per unit distance, most commonly energy per meter (E_{pm} , measured in J/m) as the main measure of energy consumption, which is a common

TABLE I

NOTATION

Parameter	Symbol	Unit
Mass of drone w/o battery	m_0	g
Mass of drone w/ battery(ies)	m	g
Drone's angle of attack	α	rad
Drone's lift-to-drag ratio	r or L/D	-
Drone's ground speed	v	m/s
Drone's airspeed	v_a	m/s
Induced speed of flowing air	v_i	m/s
Wind speed	v_w	m/s
Airspeed to headwind ratio	ϕ	-
Flight altitude	h	m
Projected area of drone	A	m ²
Drag coefficient of drone	C_D	-
Propellers' diameter	c	m
No. of blades in propeller	N	-
Blade's drag coefficient	c_d	-
Area of spinning blade disc per rotor	ς	m ²
Number of rotors	n	-
Drag force against drone	D	N
Weight force pulling drone	W or $m \cdot g$	N
Lift force of drone	L	N
Thrust force of drone	T	N
Combined force of drone	F	N
Overall power transfer efficiency	η	-
Power required for steady flight	P	W, J/s, N·m/s
Energy consumed during travel	E_{total}	J or W.s
One-way travel distance	d	m
Travel time	t	s
Air density	ρ	kg/m ³
Gravitational constant	G	N·m ² ·kg ⁻²
Gravitational acceleration	g	m/s ²

practice in delivery-focused studies. In lieu, we prefer power (in Watts) and energy (in Joules) as a function of active flight time.

Although they are consistent in their own context, many studies use their own notation for describing mathematical relations and equations, which makes it harder to conduct comparative studies and combine various models. In this sense, we appreciate the efforts of [8] on deriving a unified notation for all such models and mostly stick with it in our work with few additions. The complete list of symbols and abbreviations we use are given in Table I.

1) *Our Definition of a Small Drone*: Within the scope of this study, we are exclusively interested in small drones. As mentioned earlier, our definition of a drone is limited to electric-powered, rotary-wing multicopters, most likely a quadcopter or occasionally an octocopter. Yet we will be focusing on the earlier as the latter rarely comes in small scales but is preferred in larger designs. The lack of universal standardization and naming convention in this field of research also extends to size-based classifications. Our understanding of a "small" drone is, on the other hand, an aircraft that has a maximum diameter or diagonal length (propellers ignored) of up to 1 m, a bare (i.e., including 1 battery excluding other payloads) weight from 0.5 kg to 2 kg, and a gross (i.e., including all payloads) takeoff weight of up to 2.5 kg. Anything larger and heavier than this is out of the scope of this research, mainly due to observed differences in the energy consumption patterns of those settings and/or devices.

2) *Assumed Flight Patterns*: Most studies in the field, including this one, limit the scope of the proposed models to a so-called steady flight situation [8]. While the exact definition changes per source, the common idea is having no acceleration during the course of a flight. In lieu of that, our definition of a steady flight covers the following four cases: (i) hovering,

with zero ground speed at constant altitude; (ii) horizontal (i.e., level) flight (also called cruise), with constant ground speed at constant altitude; (iii) ascending, with zero ground speed at increasing altitude; and (iv) descending, with zero ground speed at decreasing altitude.

C. Factors Affecting Energy Consumption

There are numerous factors contributing to the overall energy consumption of a drone during its course of a steady flight (without acceleration). Examples include movement speed, wind speed, total mass, motor efficiency, air density, vessel shape, flight altitude, and energy required for avionics. Even though an ostensive classification of those factors would be internal (i.e., structural), usage-related, and external (i.e., environmental) factors, from a scientific analysis point of view, it would be a more accurate approach to evaluate the fundamental forces that an aircraft produces or counteracts to [8], [12], and [14].

1) *Fundamental Forces of a Flight*: It is an established opinion that the energy consumption of a drone during its course of a steady flight is a direct result of the forces it should produce to achieve and maintain the desired movement behavior. This includes its response and efforts to stand against the natural forces.

A drone, during a steady horizontal flight, deals with four main forces: (i) its weight, driven by gravity and pulling towards the ground; (ii) lift, the force perpendicular to the horizontal axis that it creates via its motors and propellers in order to counterbalance its weight; (iii) thrust, the propulsive force it creates again by its motors and propellers to move towards any horizontal direction; and (iv) drag, the force caused by the air surrounding the air vessel that acts against its thrust. If there is no acceleration in any direction, then the opposing forces must be equal to sustain the balance. So, the following relations were considered in the development of component-based models [10], [12]:

$$L = W \text{ and } D = T \quad (1)$$

which can also be written as;

$$\frac{L}{D} \cdot T = m \cdot g \quad (2)$$

where L , W , D , T , m , and g stand for lift, weight, drag, thrust, mass, and gravitational acceleration, respectively. Thus, the combined force that the drone should be producing to sustain its steady flight should be as follows:

$$F = W + D = L + T \quad (3)$$

Some sources, like [35], use the term thrust or total thrust (also shown as T) in lieu of the combined produced force (F), yet they refer to the lift force as thrust-to-hover (shown as T_h , T_{hover} , etc.). In order to prevent any confusion, we prefer using combined force F and the notation in Eq. 3 while doing necessary translations when referring to equations from other sources.

2) *Weight & Lift*: Weight and counterbalancing lift are functions of the mass to be flown, gravity, and air parameters,

such as air density. The total mass of a drone is a major contributor to its overall energy consumption as the need for a lift force counterbalancing and exceeding the total weight increases with it. Provided that the hardware and environmental conditions are unchanged, increasing the lift force is only possible by increasing the speed of the motors, which results in higher power consumption. The total mass includes all drone hardware, electronics, battery(ies), and additional payload, if any. It excludes components that must be kept at the ground station, like chargers, docks, cables, remote controllers, etc. We do not consider having different parameters for the mass of batteries and payloads, as doing so has little to no benefit for a parametric model.

In a balanced situation where the altitude of the drone is constant, the total forces produced by the drone to hover steadily with zero ground speed can be assumed as $F = L = m \cdot g$. As long as there is no wind, the airspeed shall be zero as well, and the lift will remain the only force that the rotors produce. Therefore, adapting from [14], the theoretical minimum power required to hover can be found using the below equation:

$$P_{hover} = \frac{L^{3/2}}{\sqrt{\frac{1}{2}\pi \cdot n \cdot c^2 \cdot \rho}} \quad (4)$$

where L , n , c , and ρ are the lift, number of propellers, propellers' diameter, and air density, respectively. If the drone is not hovering but ascending, again with a ground speed of zero, then the produced lift shall be bigger than the weight. Yet the lift will be smaller if the drone is descending. In those cases (which are revisited in later parts), airspeed and induced speed (v_i) will determine the required power.

3) *Thrust & Drag*: The thrust required at a desired cruise speed is associated with the drag. Hence, both forces are a function of the cruise speed to be maintained throughout the flight. Yet the drag is also affected by counteracting air parameters and the drone's design. While a drone's thrust must match the drag to cruise in an arbitrary direction at the desired cruise speed, both forces need to be zero to hover. The equation below shows how drag force is calculated:

$$D = \frac{1}{2} \cdot \rho \cdot v_a^2 \cdot C_D \cdot A \quad (5)$$

where ρ , C_D , v_a , and A are air density, drag coefficient, airspeed, and drone's surface area (perpendicular to the movement direction), respectively. The drag coefficient can be defined as follows:

$$C_D = \frac{2m \cdot g \cdot \tan(\alpha)}{\rho \cdot v_a^2 \cdot A_{body}} \quad (6)$$

where α is the angle of attack whose relation to the drag can be expressed as follows:

$$\alpha = \tan^{-1}\left(\frac{D}{m \cdot g}\right) \quad (7)$$

A drone's cruise speed, also called velocity, most commonly refers to its ground speed (relative to any fixed point on Earth), measured by GPS or a similar device. However, a drone's ground speed at an arbitrary time may not directly reflect its

power consumption, especially if there is significant wind. For this purpose, the (true) airspeed is a more accurate indicator as it is the vector difference between the ground and wind speeds. Wind may create a canceling force against the produced thrust or sometimes may provide a helping hand to the drone depending on its direction; hence, it may increase or decrease the power required to maintain a constant ground speed. However, having a precise real-time measurement of the wind speed may not be possible as it requires an anemometer, which is not a typical add-on for drones. If anyhow known, environmental wind speed can be reduced to headwind speed for simplicity, using the angle (θ_w) between the direction of the ground speed of the drone and the direction of the wind as $v_{hw} = v_w \cdot \sin \theta_w$. So airspeed can be obtained by a scalar subtraction $v_a = v - v_{hw}$. If θ_w is not known or is obscure due to rapidly changing flight directions, then it can be safe to assume that it is always a full headwind.

Once the airspeed is determined, the power required to maintain horizontal steady flight can be found using the conservation of momentum [14], [35] as:

$$P = F \cdot (v_a \cdot \sin \alpha + v_i); \quad (8)$$

The induced speed of the air passing through the propellers of the drone, v_i can be expressed as an implicit equation [35]:

$$v_i = \frac{2F}{\pi \cdot n \cdot c^2 \cdot \rho \cdot \sqrt{(v_a \cdot \cos \alpha)^2 + (v_a \cdot \sin \alpha + v_i)^2}} \quad (9)$$

An analytical solution for the given equation is long, but a numerical solution is easy to obtain. However, the relationship given in eq.(9) can only be assumed in close-to-ideal settings where there is no significant wind or other external forces. For example, [35] states that the accuracy of the above equations will be lower for larger angles of attack and will be invalid for the vortex ring state. Please note that Eq. 4 stating the power to hover is a special case of Eq. 8 and can be derived from Eq. 8 and Eq. 9, by setting airspeed v_a to zero and F to L .

4) *Wind Speed*: In general, wind can have both positive and negative effects on a drone's energy efficiency, depending on the direction and speed of the wind. For example, a headwind can slow down the drone and increase its energy consumption if higher speeds are intended, while a tailwind can speed up the drone and decrease its energy consumption. Since there is a setpoint for the ground speed in most cases, on the other hand, extreme tailwinds that would otherwise force the drone to overspeed may also cause increased energy consumption as the drone must apply a counter force (analogous to braking) to maintain its speed per the setpoint. Crosswinds can also affect the drone's stability and maneuverability, which may require additional energy to control. The impact of crosswinds may be as bad as headwinds [15].

At lower altitudes, particularly below the planetary boundary layer, wind speed tends to be slower due to surface (i.e., terrain) friction and natural or man-made obstacles. The extent of the planetary boundary layer also depends on the time of the day and the temperature, yet estimates start from 100 m to 457 m for large cities, 366 m for suburbs, 274 m for open terrain, and 213 m for open sea [36]. Above the planetary

boundary, such as around 1000 m, aircraft may face a more powerful and more steady gradient (e.g., geostrophic) wind.

Small drones are mostly designed to be used in lower altitudes; hence, are subject to varying rates of (hourly mean) wind speed, calculated using the power-law relation [36]:

$$v_w(h) = v_w(h_{ref}) \cdot \left(\frac{h}{h_{ref}} \right)^\gamma \quad (10)$$

where $v_w(z)$ is the hourly mean wind speed at height h above the ground, h_{ref} is the reference height for the observed wind (typically 1.5 m or 10 m), and γ is the exponent (defined as 0.33 at city centers, 0.22 at suburban areas, 0.14 at open terrain, and 0.1 at open sea).

The flight altitude alone may not be a major contributor to the power requirements in certain conditions but affects it indirectly as the wind speed and air density may change according to the altitude. The effects of wind speed were already discussed above. The air density also decreases with altitude. As air density drops, the motors should run at higher rates to maintain the required lift. Nevertheless, this effect is minimal at the heights where a small drone is expected to operate and can be neglected.

5) *Motors & Power Efficiency*: The energy that is being transferred from the source (i.e., battery) to the motors is subject to various losses (e.g., heat) during each stage of the transfer, yet the motors also have a loss rate for the conversion of electrical energy to kinetic energy that rotates the propellers. This entire chain of losses yields a unitless power efficiency ratio that should be multiplied by the required power to fly calculated from the required forces. The ratio can be applied as follows:

$$P_{\text{eff}} = P \cdot \eta \quad (11)$$

where P_{eff} , P , and η are the effective power that shall be drawn from the battery to make up for the inefficiency, the actual power required for movement, and the power efficiency ratio, respectively. Note that the efficiency may not be increased by simply having more powerful motors as there is also the impact of the motors' own mass, which is usually higher with more powerful ones.

The energy required to "recharge" a battery is also larger than the energy that can be stored in the battery due to losses occurring during charging and waiting on the shelf. However, what we consider as the energy cost of a swarm operation is not the cost of electrical energy transferred from the grid; instead, we focus on the lifetime of the drones, hence considering the efficiency from the battery to the motors only. Readers may refer to [14] for battery efficiency concerns.

6) *Avionics*: Avionics is a portmanteau of aviation electronics and mainly refers to all computation and communication devices equipped by the subject drone during its flight. Although there is no limitation on what could be among such devices, typical examples include a flight controller (e.g., PX4, ArduPilot, etc.), a flight computer (e.g., ModalAI VOXL, Nvidia Jetson, RaspberryPi, etc.), and other processing units like digital signal processors (DSP), cameras, radio controllers (e.g., WiFi, Microhard, etc.). Extensive use of processors (e.g., for edge computing) or add-on devices that require a lot

TABLE II
EXPERIMENT PARAMETERS

Parameter	Symbol	Value	Unit
Mass of drone w/o battery)	m_0	786	g
Mass of drone w/ 1 battery	m_1	1102	g
Mass of drone w/ 2 batteries	m_2	1418	g
Mass of drone w/ 3 batteries	m_3	1734	g
Projected area of drone	A	0.0599	m^2
Drag coefficient of drone	C_D	-	-
Propellers' diameter		0.254	m
Number of rotors	n	4	-
Overall power efficiency	η	-	-
Air density	ρ	1.225	kg/m^3
Gravitational constant	G	6.674×10^{-11}	$N \cdot m^2 \cdot kg^{-2}$
Gravitational acceleration	g	9.807	m/s^2

of power (e.g., infrared sensors) would cause those devices to dominate the overall energy consumption; however, those cases are evaluated as exceptional and are ignored in this work. Yet, [22] suggests that non-special computation and communication components are roughly responsible for 5% of the total energy consumption of a flying drone.

IV. EXPERIMENTS

A. Flight Scenarios

Distinct flight tests were conducted for steady horizontal cruise, hovering, and vertical movements. However, vertical movements were later excluded due to our drone's incapability of reaching higher speeds vertically. All test flights were performed at the San Diego State University's main campus and in the Chollas Lake Park area, both in San Diego, California.

B. Testbed Drone Hardware

The testbed used throughout the experiments was a ModalAI m500, a commercially available programmable drone equipped with a WiFi transceiver, a Spektrum radio receiver, and a GPS module. The ModalAI m500 developer drone is built on the well-established S500 frame and Qualcomm RB5 platform. It accommodates a PX4 flight controller and a VOXL flight companion computer. The flight controller works actively and near full load during any flight, yet the VOXL companion computer works under minimal load (due to some background services like the MAVLink server) unless otherwise programmed. The drone and its dimensions can be seen in Fig. 1.

The motors installed are 4× Holybro 2216/880kv. The battery used is a ProLiteX 4-cell 25 C lithium-polymer (Li-Po) battery. Additional batteries of the same kind are used to demonstrate a payload-carrying scenario. While a fully configured m500 drone without any battery installed weighs 786 g, it weighs 1102 g with a single battery, 1418 g with double batteries (one for payload), and 1734 g with three batteries (two for payload). In our setting, those additional batteries were not connected to the power line but were used as payload only. For more information about hardware, the datasheet for the motors used with the drone can be found online (<https://github.com/cbk35/DroneEnergy/blob/main/2216-880kv.jpg>). The origin of the power-related measurements found in the logs is indeed the main power module used in the drone. The datasheet of the main power module used in the drone can

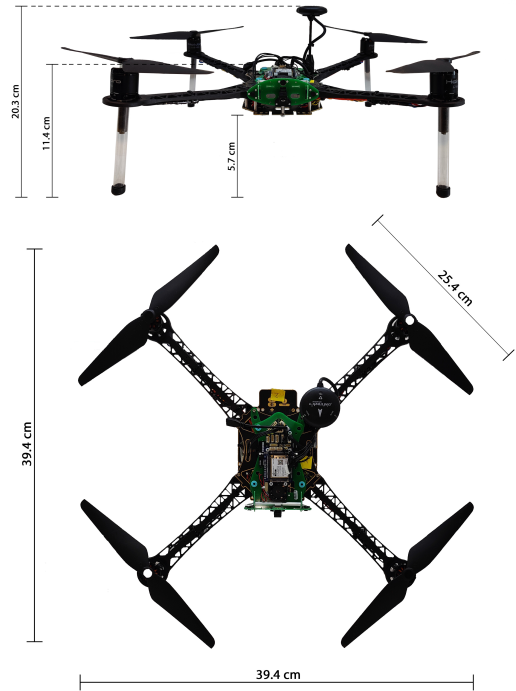


Fig. 1. ModalAI m500 testbed drone and its dimensions.

be accessed online (<http://docs.modalai.com/power-module-v2-datasheet>).

C. Measurement Methods and Dataset

Each flight scenario defined in Section IV-A was uploaded to the drone's flight controller (i.e., PX4) using the QGround-Control program through a WiFi connection, and the entire course of the flights is tracked through the same program (except the rare cases where WiFi connection was lost). The process is repeated until all scenarios are realized. The drone records a lot of information during the flight as time series data in "ulg" log files. The logs were analyzed and visualized using the PlotJuggler program. Readings were done manually using charts.

Our flight dataset is entirely built by us through real flight experiments made on the field. The set consists of a total of 98 data entries (i.e., rows), each belonging to one full flight with the specified parameters. In particular, there are 45 entries for horizontal steady flight, 22 entries for hovering, 9 entries for ascending movement, and 8 entries for descending movement. Apart from identifying information, there are four features (i.e., columns): wind speed (m/s), ground speed (m/s), flight altitude (m), and total mass (g), whereas there is one target variable: the power drawn during the action (W), calculated using the battery discharge (mA·h) in a given time window (s) and the average voltage (V) in the specified window.

1) *Wind Speed Adjustment*: Data regarding the speed of wind at the time of the flights were not collected by an onboard anemometer but obtained from an online database keeping the records of weather data measured at San Diego International Airport, which is circa 10 km away from both experiment locations. Data is available at (<http://www.wunderground.com/history/daily/us/ca/san-diego/KSAN>). National Oceanic and

Atmospheric Administration (NOAA) reports that the measurement height for wind speed is 1.5 m above the surface; however, it is subject to change per altitude as discussed in Section III-C3. Therefore, the measured wind speed should be adjusted according to the actual flight altitudes using Eq. 10, for better accuracy. The assumed flight environment was suburban, hence the coefficient of 0.22. This new feature is called adjusted wind speed (m/s) and entirely replaced the original wind speed feature. In fact, we have repeated our model-building efforts using the original wind speed and the adjusted wind speed, yet all other things being equal, the adjusted wind speed provided better results with all four metrics introduced in Section VI. Nevertheless, the altitude feature is not removed to keep potential correlations with other features. Other data preprocessing techniques applied to the dataset are introduced in Sections V-C1 and V-C2.

V. MODELING ENERGY CONSUMPTION

The energy consumption of a drone is modeled using three different approaches: theoretical analysis, multivariate linear regression (both simple and polynomial), and machine learning via XGBoost (with data augmentation and autoencoding).

A. Theoretical Model

Our theoretical model is mainly derived from the equations Eq. 4 Eq. 8 Eq. 3 Eq. 10 introduced earlier. For simplicity, parameters that are harder for a typical drone operator or mission planner to obtain in real-time can be replaced with an observed or pre-calculated constant. Since we are not aiming at a best-case estimation (in fact, it is mostly the opposite), it is convenient to substitute some variables with their average-to-worst-case values.

The area swept by all four propeller blades is pre-calculated as $A = 0.203 \text{ m}^2$ from the dimensions given in Fig. 1. Likewise, ρ and g have default values given in Table II. Those values are typical for small drones operating on low altitudes; hence, small changes will not have a significant impact. From the experiments, the maximum angle of attack (α) observed during a horizontal flight, which is equal to the pitch angle when there is no wind, was 41.6° with a ground speed (v) of 19 m/s and a wind speed (whose direction was obscure) of 5.2 m/s. Taking this as an extreme point, an angle of 30° is presumed. So we make the following substitutions $\sin \alpha = 0.5$, $\cos \alpha = 0.866$, which altogether yields the intermediate formulas for the required power $P = F \cdot (v_a \cdot 0.5 + v_i)$ and for the induced velocity v_i :

$$v_i = \frac{2F}{0.2482 \cdot \sqrt{((v_a) \cdot 0.866)^2 + ((v_a) \cdot 0.5 + v_i)^2}} \quad (12)$$

Furthermore, v_i can be reduced to a polynomial of the third degree via numerical calculation of its values per airspeed (v_a), where v_a varies from 0 to 25 m/s. Thereafter, it is fitted to a cubic polynomial using regression. On the other hand, quite optimistically, F can be reduced to the lift (L) using $m \cdot g$, where $m = m_1$. Note that the datasheet gives F in gram-force, which needs to be converted to Newton first. Hence,

the resulting equation for v_i as a function of m and v_a can be obtained as below ($R^2 = 0.9999$):

$$v_i = -0.0002131 \cdot v_a^3 + 0.01722 \cdot v_a^2 - 0.5592 \cdot v_a + 9.595 \quad (13)$$

By substituting the above polynomial for v_i in the equation of P , we can get a clearer formula for the required power. However, those relations do not consider the energy efficiency of the motors. Thus, we need to find the efficiency factor first. Although the motors' power efficiency is not constant as it changes with angular speed and torque generated, we presume it is constant, which is also a common practice in the literature [14]. According to the seven data points provided in the datasheet, the power efficiency varies from 74.93% to 80.15% for various throttle levels. In order to prevent higher speeds from being dominant and for simplicity, we accept the efficiency rate as 80%. The power efficiency factor (η) is $1/(\text{efficiency rate})$ and should be multiplied by the required power anticipated so far. To sum up, the power consumed by the drone can finally be obtained as follows:

$$P_{\text{final}} = 1.25 \cdot m \cdot g \cdot (v_a \cdot 0.5 + (-0.0002131 \cdot v_a^3 + 0.01722 \cdot v_a^2 - 0.5592 \cdot v_a + 9.595)) \quad (14)$$

On the other hand, the relations obtained until this point exclude the impact of avionics included in the drone. As suggested by [22], an additional adjustment of up to +5% can be added to include the power for avionics (by multiplying the computed formula for P by 1.05) if there is significant use of avionic devices, such as the radio communication systems, camera, or processing units. But note that the effects of avionics may disappear at higher speeds or weights. Our experiments did not involve such uses; thus, we excluded it.

Since we have narrowly limited our definition of a small drone, minor differences in parameters may only affect the accuracy of the model slightly. Nevertheless, having a theoretical model in hand is very useful for understanding the factors that may have a significant effect on overall energy consumption. However, such models are only good as an overview because they are not as accurate as they must be to be relied on for calculating the actual energy cost of flights, especially for swarms where smaller errors would have a snowball effect. In fact, it is also stated by others [14] that those models are good for obtaining the theoretical minimum, whereas, in practice, real consumption values are expected to be higher than what the theoretical models suggest.

B. Regression Model

Multivariate linear regression is a neat way to derive a single equation for energy consumption calculations that employ all the factors. While a simple linear regression can provide a quick but less accurate linear relation between the dependent and independent variables, a polynomial linear regression, on the other hand, can provide a more accurate polynomial relation, as it employs power terms and even interaction terms (i.e., terms combining multiple factors). Please note that the linearity in polynomial regression is preserved on the coefficients, not on the variables.

We computed both models and found the best coefficients that minimize the errors. Both regression models are built using experimental data and reflect the impact of all subsystems of a drone, including avionics. Yet, the polynomial model is much more accurate according to all metrics. Power with simple linear regression is given below:

$$P = \beta_0 \cdot v + \beta_1 \cdot w + \beta_2 \cdot h + \beta_3 \cdot m \quad (15)$$

The computed coefficients are given below:

$$\begin{aligned} \beta_0 &= 3.61731127 & \beta_1 &= 0.15371393 \\ \beta_2 &= -0.11874969 & \beta_3 &= 8.85335471 \end{aligned} \quad (16)$$

Power with polynomial linear regression is given below:

$$\begin{aligned} P &= \beta_0 + \beta_1 \cdot m + \beta_2 \cdot v^2 + \beta_3 \cdot v \cdot m + \beta_4 \cdot m^2 \\ &+ \beta_5 \cdot v^3 + \beta_6 \cdot v^2 \cdot m + \beta_7 \cdot v^2 \cdot v_w + \beta_8 \cdot v \cdot m^2 \\ &+ \beta_9 \cdot m^3 + \beta_{10} \cdot m^2 \cdot v_w \end{aligned} \quad (17)$$

The computed coefficients are given below:

$$\begin{aligned} \beta_0 &= 91.3914 & \beta_1 &= 1.25840562e - 08 \\ \beta_2 &= -2.1809981e - 01 & \beta_3 &= -1.77253268e - 03 \\ \beta_4 &= 1.75488067e - 05 & \beta_5 &= 4.48969682e - 02 \\ \beta_6 &= -3.37373692e - 04 & \beta_7 &= 3.87283483e - 02 \\ \beta_8 &= 9.96433977e - 07 & \beta_9 &= 2.44381779e - 08 \\ \beta_{10} &= -3.28137684e - 06 \end{aligned} \quad (18)$$

The equation obtained by the simple linear regression performed poorly in terms of estimation accuracy and should not be relied on for serious calculations; the equation obtained by the polynomial regression is, however, significantly more accurate (in terms of all our metrics introduced later in Section VI-A) and can be used in applications where small to medium variance is tolerable. Apparently, other solutions should be sought for use in high-precision scenarios.

C. Machine Learning-Based Model

Considering the multivariate nature of the experimental flight data (and how theoretical models underestimate energy consumption), we have anticipated that a machine learning model can master the relations between the variables and provide a more accurate way of predicting prospective energy consumption in various flight scenarios.

For this purpose, XGBoost (extreme gradient boosting) is considered a viable machine learning and prediction method. It is a supervised learning method capable of making regression for time-series-like data and classification for labeled data [37]. Our flight data is organized entirely as time-series data and is provided as input to the XGBoost's regressor. Note that, like linear regression, the machine learning-based model also reflects the impacts of all components and avionics.

The entire source code of the study (including training, testing, estimating, and reporting functions) can be found at (github.com/cbk35/DroneEnergy/blob/main/EnergyModel.ipynb), as a Jupyter Python Notebook. The entire flight dataset, as well as a pre-trained model file to be used in further research, is also available from the authors upon reasonable request.

1) *Data Augmentation*: Our dataset is relatively small, and our model might have been impacted negatively due to this fact, which later turned out to be true. In order to mitigate the potential issues caused by a smaller dataset, such as low accuracy and high error, the dataset should be expanded. Yet, we did not consider expanding our dataset through synthetic data generation (i.e., imputation) methods that perform an “educated” guess on so-called missing data points (e.g., datawig) because such methods would execute a preliminary round of machine learning-based fitting before the actual model's fitting takes place. This behavior is, most likely, not beneficial for the actual model as it obscures the organic relationships between the features and the observed variable. We note that SMOGN [38], the time series version of SMOTE, is not useful either, as it causes losses in the original dataset.

Instead, we applied data augmentation to our dataset by including “noisy” variations of each real measurement data. So that each data point is precisely replicated 10 times, and a randomly generated Gaussian white noise with zero mean of a $\pm 2\%$ margin was applied to each replicated entry. In this way, we could improve the size, balance, and variety of our dataset without distorting the organic correlations.

2) *Autoencoder & Normalizer*: Since the flight dataset contains multiple features, each varying in different ranges, it is anticipated that applying normalization can improve the machine learning procedures. What is also thought to be beneficial in this sense is running an autoencoder that manipulates the original data by passing it through a few layers of a neural network to signify the most impactful features. Hence, we applied those two measures before proceeding with the actual machine learning procedure.

We have used the autoencoder introduced in Python's Keras library with 6 dense layers included. The encoder is trained on the normalized training data using the mean squared error as its loss function and utilized the Adam optimizer [39] for 100 epochs with a batch size of 32. The autoencoder model is then used to obtain encoded data from the training and test data. Especially when following a data augmentation step, the data normalization and autoencoding steps prior to the machine learning phase largely increase the accuracy and performance of the resulting machine learning model built via XGBoost.

D. Cost Function for Swarm Topology Control

The total (i.e., combined) energy consumption of a drone swarm during an observed time frame provides a useful basis to compare various activities that the drones may be performing in the same time frame; thus, the most feasible option can be determined. This is especially useful for evaluating various connectivity restoration, topology setup, and formation-changing methods. Such a notion of energy cost can best be considered by assuming that the swarm is not operable (in the sense of servicing within its specific mission) when the wireless network it otherwise keeps active is disconnected. In this case, the energy cost must be measured until the connectivity is restored. Likewise, the energy cost of recreating the topology, as well as that of changing the flight formation,

TABLE III
ESTIMATION PERFORMANCES OF DEVELOPED ENERGY MODELS WITH VARIOUS SETTINGS

Model	Contextual Preproc.	Data Augmentation		Accuracy Metrics			
	Wind Adjustment	AWGN Reproduction	Autoencoder & Norm.	MSE	RMSE	MAE	R ²
Theoretical Model	N/A	No	N/A	1053.121	32.452	18.145	0.557
	N/A	Yes	N/A	719.317	26.820	16.021	0.719
Simple Linear Reg.	No	No	N/A	673.926	25.960	18.641	0.745
	No	Yes	N/A	674.767	25.976	18.665	0.745
	Yes	Yes	N/A	703.920	26.531	18.404	0.734
Polynom. Lin. Reg. (Cubic)	No	No	N/A	89.344	9.452	7.087	0.967
	No	Yes	N/A	93.308	9.660	7.264	0.965
	Yes	Yes	N/A	88.948	9.431	7.155	0.966
ML: XGBoost Regressor	No	No	No	290.134	17.033	11.433	0.878
	Yes	No	No	250.526	15.828	10.423	0.895
	No	No	Yes	85.456	9.244	6.655	0.964
	No	Yes	No	13.466	3.670	2.415	0.995
	No	Yes	Yes	0.313	0.559	0.380	0.9999
	Yes	Yes	Yes	0.222	0.471	0.301	0.9999

can be measured from the starting time of the process until the desired setting has been reached, even if no disconnection occurs during the process at all.

As a cost function, we derived the following equation concerning the total energy consumed by the entire swarm of n drones, exclusively during the connectivity restoration, formation change, or topology maintenance processes (E_{main}):

$$E_{main} = n_{idle} \cdot P \cdot t_{total} + P_1 \cdot t_1 + \dots + P_m \cdot t_m \quad (19)$$

where n_{idle} is the number of drones waiting idly (e.g., by hovering) until connectivity is reestablished or topology maintenance is successfully performed, excluding the ones obliged to relocate to execute the topology maintenance ($n_{idle} < n$). P_i is the power requirement of each drone that must be relocated during maintenance. As they may have different conditions (e.g., ground speed, weight, etc.), their contribution to overall consumption should be considered separately. Last but not least, t_{total} is the total time required for the entire topology maintenance process to converge, yet t_i is the movement time of each drone obliged to relocate ($t_n < t_{total}$).

It is clear from the cost function given in Eq. 19 that optimizing energy consumption requires minimizing total time t_{total} . Given $t_{total} \leq \sum_{i=1}^m t_i$ (equality is the worst case), minimizing t_{total} requires (i) relocating drones simultaneously, (ii) relocating as many drones as possible, (iii) allocating travel distances to each relocating drone to satisfy $\max(t_1, \dots, t_m) \approx \bar{t}_i$, and (iv) consider moving at higher speeds, especially when $n_{idle} > m$. Moreover, the cost function implies that the assumed topology formation of the swarm does not have an immediate impact on energy consumption in an ad-hoc network representing a unit disk graph. If the link lengths are not uniform, then a formation with smaller inter-node distances has a fair edge on energy efficiency as it has shorter t_i in general.

VI. RESULTS AND DISCUSSION

This section compares all four energy models that we have come up with and discusses their estimation accuracy, strengths, and limitations.

A. Model Accuracy

For quantifying the model accuracy, we used mean square error (MSE), root mean square error (RMSE), mean absolute error (MAE), and coefficient of determination (R^2) against our experimental flight data. For the comparison charts presented in this section, we preferred using the required power (in W) as the target variable for convenience. Readers may find the energy consumption (in J or W·h) in a specific time frame by multiplying the given power values (in W) with the duration (in s or h) of their wish.

Table III constitutes our main reference for the comparison of all four models with different configurations. In fact, it is pruned by eliminating some statistically less meaningful configurations and made more concise. Yet, not all the configurations are possible for all the models. There are three major data configurations considered: one related to the so-called contextual preprocessing of our data, which is adjusting the wind speed per Eq. 10, and two data augmentation techniques, which are AWGN reproduction and an autoencoder mechanism following range normalization. Wind speed adjustment is not applicable to our theoretical model as it does not have such a parameter. While AWGN reproduction of data is applicable to all models, autoencoder with normalizer is only applicable to our ML-based model.

Our evaluation procedure involves estimating the power consumption values (in W) as a series per a set of given parameters, namely, ground speed (in m/s), gross weight (in g or kg), wind speed (in m/s), altitude (in m) using each model. The parameter combinations follow the ones of our real experimental data. Finally, the abovementioned performance metrics were calculated. Per the results, possibly since our experimental dataset is relatively small, artificial reproduction of new data entries by repeating existing ones with a small AWGN variation improves the accuracy of the models very significantly, except for the simple linear regression, where it does not have any effect. Adjusting the wind speed by calculating the equivalent wind per flight altitude also increases the accuracy noticeably, though not largely. The autoencoder is found to be useful for our ML model as well.

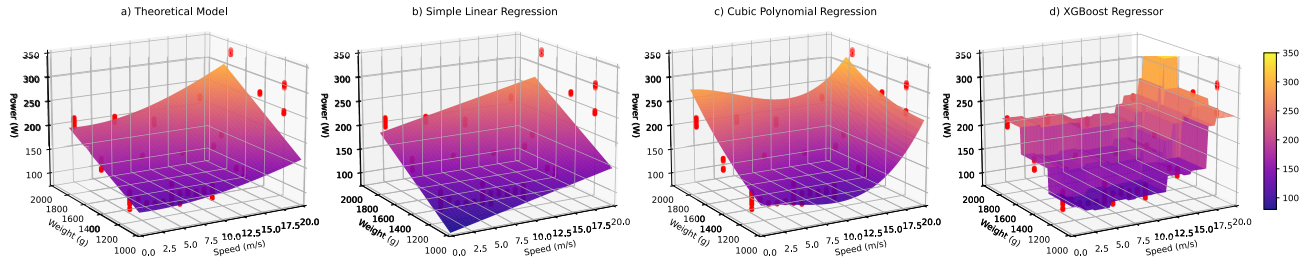


Fig. 2. Power projections of the created models per varying speed and weight.

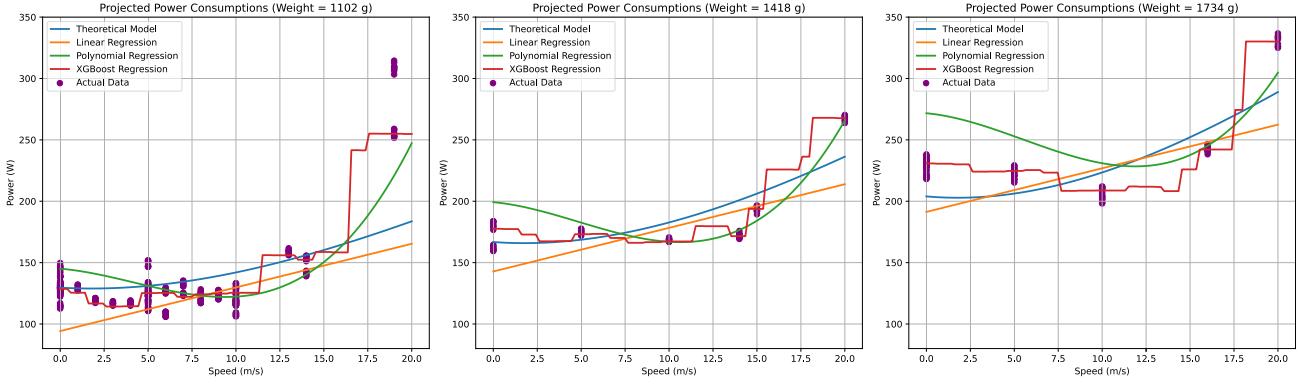


Fig. 3. Power projections of the created models per varying speed and weight.

As clearly presented in Table III, considering the best configurations for any given model, the XGBoost regressor provides the highest accuracy among all with an MAE of 0.3 (within the true value range of 80 to 350 Watts) and an R^2 of 0.9999, which is extremely accurate. It is followed by the cubic polynomial regression, which also has a very high accuracy with an MAE of 7.2 (in the same range) and an R^2 of 0.966. The theoretical model and the simple linear regression were fairly accurate in some cases and not that much in others, with MAEs of 16 and 18.4, R^2 s of 0.719 and 0.734, respectively. The theoretical model, though becoming a lot better with AWGN data augmentation, lacks accuracy. Hence, projections based on the theoretical model should not be relied on when quick and coarse estimations are acceptable. The simple linear regression model is at least as accurate as the theoretical model and is easier to implement, but it misses important non-linear characteristics; hence, it may find its use among several, mostly non-scientific, scenarios. Fig. 2 shows how well (or poorly) the models fit the experimental data, which is consistent with the metrics given in Table III.

B. Energy Consumption Projections

In order to assess the usability of the models, various power requirement projections were made aiming at horizontal flights with varying speeds and with different weights.

Fig. 2 presents comprehensive 3-D surface drawings illustrating the power projections obtained by each created model per varying speed and weight values. Red dots indicate experimental measurements at various wind levels, while surfaces consider zero wind conditions. This analysis does not include vertical movements as well. The digital version of this paper allows to zoom in for a more detailed view. The speed range is between 0 m/s (i.e., hovering) to 20 m/s, while the weight range is between 1000 g to 2000 g, while wind was zero.

According to the plots, the simple linear regression and the theoretical model fall behind when reflecting the effect of changes in speed and weight on the power drawn. They also mostly underestimate the power (per experimental data) for larger values of speed and weight. The polynomial regression and the ML-based model are visibly prompter and more precise when reflecting the effect of changes in speed and weight, whereas the latter is finer. On the downside, however, the ML-based model, due to its decision-tree-based nature, cannot be used in extrapolating projections outside of its training range. Though interpolations are not natively supported as well, it is fairly easy to make projections using linear fittings between training points. Nevertheless, polynomial regression is viable for interpolations and extrapolations to a greater extent.

Fig. 3 provides a basis for a more concrete comparison as it gives power values at speeds varying from 0 to 20 m/s while weight, altitude (20 m), and wind speed (0 m/s) parameters were kept constant. Three subplots represent repeated projections at different total weights, at 1102 g, 1418 g, and 1734 g, respectively.

All models react to the increase in speed or weight in the right direction; however, their accuracy is different, as discussed earlier. Though non-existent in simple linear regression and barely noticeable in theoretical model, polynomial regression and ML-based models unveil an efficient range of horizontal speeds that represent a “valley” in the power charts. This “valley” indicates that moving at certain, so-called mediocre speeds requires less power than hovering still, moving at very slow speeds, and also moving at much higher speeds. According to polynomial regression and ML-based models, the deepest point of that “valley” roughly hits 6 m/s for 1102 g, 10 m/s for 1418 g, and 12.5 m/s for 1734 g, where the required power is found around 130 W, 170 W, and 220 W, respectively (averaged from both models). The

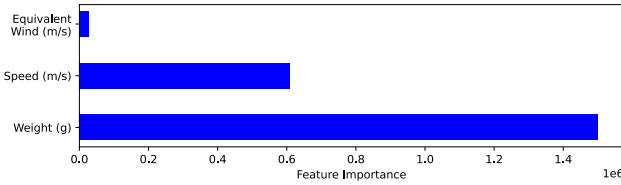


Fig. 4. Importance of features in XGBoost model for energy.

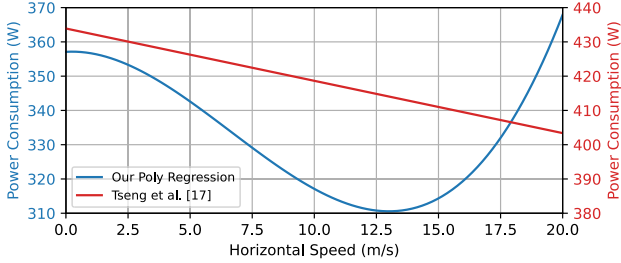


Fig. 5. Comparison of our regression model and the one in Tseng et al. [17] emphasizing the energy-efficient “valley”.

power to hover is 140 W, 185 W, yet the maximum power, consistently at 20 m/s, is around 250 W, 270 W, and 330 W for 1102 g, 1418 g, and 1734 g, respectively. Note that experimental data shown by purple dots in Fig. 3 include measurements at different wind speeds up to 5 m/s, hence the vertical trailing effect (due to the lack of flight data for no wind conditions).

To visualize how instrumental this energy-efficient valley is in defining energy consumption characteristics of drones, we present another comparison, between our polynomial regression model and the regression model (for small drones) proposed in [17] (and later used in [21]). In constructing that model, flights were conducted at horizontal speeds ranging from 0 to 5 m/s for small drones (3DR Solo) and 0 to 12.5 m/s for large drones (DJI M600). We, on the other hand, covered speeds from 0 to 20 m/s. Depicted in Fig. 5 is a simplistic scenario with no wind, no horizontal or vertical acceleration, constant altitude, and a total weight of 2 kg. The model in [17] yields an ever-decreasing power demand projection as horizontal speed increases. While this is consistent with the quasi-linear downward slope observed in our model, after the sweet spot encountered around 13 m/s (subject to change per weight) is achieved, our power projection starts indicating a dramatic increase in demand.

C. Limitations of Study

Although their impacts are found to be acceptable, there are a few limitations throughout the study worth noting:

- The wind data used was not measured on the drone but obtained from an online database. The closest reporting station was 10 km away from our field, and data was being updated hourly; yet, the flights involved various directions with no consideration of the actual angle between the heading and the wind direction, which may have reduced the accuracy of wind impact (see Fig. 4).
- Our ML method, XGBoost, does not allow extrapolations beyond or measured data range. Hence, while interpolations are less of a concern as data points were sufficient, generalization of the ML-based model out of the data

ranges is not viable. In such cases, a fallback on the polynomial regression may work.

- The only drone used was a quadcopter; so, all models and formulations given are merely valid for drones alike but not for hexacopters, octocopters, fixed-wings, etc.
- Due to size constraints of authorized flight zones, it was impossible to continue high-speed flights (e.g., 20 m/s) for longer than a minute, limiting data accuracy.

D. Key Takeaways

The most notable findings of the study are summarized:

- There is an energy-efficient “valley” within the airspeed spectrum (see Fig. 2), which is roughly between 4 to 12 m/s for a small drone with no payload. Flying at speeds out of this range costs more energy per time and distance.
- The energy-efficient “valley” moves towards faster speeds as the carried weight increases. So, flying at higher speeds is more feasible when there is more weight to carry.
- Hovering costs more energy than a steady flight at a constant speed within that energy-efficient “valley”.
- The impact of the flight altitude on energy consumption is very small in the range of 0 to 100 m from the ground level. Yet, at higher altitudes, it may be observable.
- Our XGBoost Regressor-based machine learning model is, by far, the most accurate one among all four options; however, its accuracy does not extend over the values not covered by the training data. For extrapolation purposes, the polynomial regression-based model provides the best performance. In comparison, the theoretical model tends to underestimate energy consumption at higher speeds.

Last but not least, Fig. 4 provides an insight into the importance of each feature included in our ML-based model. The importance metric was total gain, and values were found after a grid search aiming at determining the best parameters. The figure illustrates how important a drone’s weight is.

VII. CONCLUSION

In this work, we focused on the problem of accurately estimating the energy consumption patterns of small drones that weigh less than 2 kg and are smaller than 1m in diameter. We have made numerous test flights and collected experimental energy consumption data. Subsequently, we built and proposed four distinct energy models: a theoretical model based on the calculation of fundamental forces of flight, a simple linear regression model fitting our experimental data, a cubic polynomial regression model again fitting experimental data, and lastly, a machine learning-based model utilizing XGBoost Regressor algorithm on the experimental data.

While all models may be preferable in various use cases, our ML-based model is extremely accurate (with an R^2 of 0.9999) in the range that it was trained. Outside the range, however, it is less useful. For extrapolating the dependent variables to our polynomial regression-based model is the way to go with decent accuracy (with an R^2 of 0.966). Though less accurate, the linear regression model can be a simple tool to quickly calculate the energy requirements, whereas suffering from the

same accuracy concerns, the theoretical model can be another option, especially when modifying the hardware parameters. Lastly, a model-independent generalization of the collective energy consumption of a drone swarm is formulated, focusing on topology maintenance scenarios where some drones relocate while others wait idly.

Future works will involve further analysis of the impact of wind speed and direction on energy consumption using an onboard anemometer, as well as power requirements at various acceleration levels, which is impacting energy consumption significantly, according to our observations. We also aim to extend our model to cover even smaller “micro” drones.

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